MIL-M-28787/128B(EC)
4 May 1979
SUPERSEDING
MIL-M-28787/128A(EC)
27 January 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TRANSFORMER, ANALOG, KEY CODE CTR

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This document establishes the requirements and test procedures for the procurement of an electronic transformer module

1 2 Module description The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I

Name	Transformer
Family	Analog
Key code	CTR
Size	1E
Weight	454 grams maximum
Failure rate	0 983 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	10 mW
Rated input voltage	30 Vrms, 9 6 kHz
Maximum working voltage	85 Vp-p
Transformer ratios (primary to secondary)	1 1 ±0 25 percent
Primary	
Tap 1	1 0 975 ±0 125 percent
Tap 2	1 0 950 ±0 125 percent
Secondary	
Tap 1	1 0 500 ±0 125 percent
Tap 2	1 0 284 ±0 125 percent
Tap 3	1 0 229 ±0 125 percent
Tap 4	1 0 179 ±0 125 percent
Tap 5	1 0 100 ±0 125 percent
Tap 6	1 0 0179 ±0 50 percent
Tap 7	1 0 0156 ±0 50 percent
Phase shift	0 ±0 25 degree
Harmonic distortion	1 percent maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, D C 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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Frequency response	±3 dB, 400 Hz to 100 kHz
DC resistance	
Primary	41 09 ohms maximum
Secondary	41 09 ohms maximum
Continuous load power	0.025 W per transformer

1.3 Module function. This module contains two general purpose transformers which are identical but electrically isolated. Each transformer has one primary with two taps and one secondary with seven taps. There is an electrostatic shield between the primary and the secondary of each transformer.

2 APPLICABLE DOCUMENTS

2 1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

3 REQUIREMENTS

3 1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein

3 2 Correlation requirements Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

3 3 Failure rate requirements All modules procured to this specification shall have a predicted failure rate no greater than 0.983 failures per million operating hours when predicted in accordance with MIL-STD-1389

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389

3 5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3 5 1 Key code. The module key code shall be CTR

3 5.2 Size. The module size shall be 1E

3 5.3 Weight The module shall have a maximum weight of 454 grams

3.5 4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5.6 Depth The module shall have no additional depth requirements

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TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature Initial and end-of-life		Operating temperature 0° to 60°C Initial and end-of-life		Unit
				Min	Max	Min	Max	
A1-A4	Transformer ratio (full primary to full secondary)	II	X	1.0 9975	1 1 0025	1 0 9975	1 1 0025	Ratio
B1,B2, B5,B6	Tap accuracy (primary to primary), tap 1	III	X	1 0 974025	1 0 975975	1 0 973781	1 0 976219	Ratio
B3,B4, B7,B8	Tap accuracy (primary to primary), tap 2	III	X	1 0 94905	1 0 95095	1 0 948813	1 0 951188	Ratio
B9-B12	Tap accuracy (secondary to secondary), tap 1	III	X	1 0 4995	1 0 5005	1 0 499375	1 0 500625	Ratio
B13-B16	Tap accuracy (secondary to secondary), tap 2	III	X	1 0 283716	1 0 284284	1 0 283645	1 0 284355	Ratio
B17-B20	Tap accuracy (secondary to secondary), tap 3	III	X	1 0 228771	1 0 229229	1 0 228714	1 0 229286	Ratio
B21-B24	Tap accuracy (secondary to secondary), tap 4	III	X	1 0 178821	1 0 179179	1 0 178776	1 0 179224	Ratio
B25-B28	Tap accuracy (secondary to secondary), tap 5	III	X	1 0 0999	1 0 1001	1 0 099875	1 0 100125	Ratio
B29-B32	Tap accuracy (secondary to secondary), tap 6	III	X	1 0 017855	1 0 017945	1 0 017811	1 0 017990	Ratio
B33-B36	Tap accuracy (secondary to secondary), tap 7	III	X	1 0 015561	1 0 015639	1 0 015522	1 0 015678	Ratio
C1-C4	Phase shift (full primary to full secondary)	IV	X	-0 25	+0 25	-0 25	+0 25	Degree
D1,D2	Leakage inductance (reflected to secondary)	V	S	0	0 003	0	0 003	H
E1-E4	Primary open circuit resistance and reactance	VI	S	80,000 +j80,000	---	80,000 +j80,000	---	Ω
F1,F2	Capacitance (primary to secondary)	VII	S	0	50	0	50	pF
G1-G4	DC resistance (primary and secondary)	VIII	X	18	33	---	41 09	Ω
H1-H4	Total harmonic distortion	IX	S	0	1 0	0	1 0	Percent

See notes at end of table

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TABLE I Module electrical requirements -Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature Initial and end-of-life		Operating temperature 0° to 60°C Initial and end-of-life		Unit
				Min	Max	Min	Max	
J1,J2	Frequency response (400 Hz to 100 kHz)	X	S	-3	+3	-3	+3	dB
K1,K2	Electromagnetic coupling	XI	S	0	0.1	0	0 1	Vrms
L1-L11	Integrity	XII	S					
	Initial			10	∞	---	---	MΩ
	End-of-life			1 0	∞	---	---	MΩ
M	Continuity	XIII	S	0	0.1	0	0 1	Ω
N	Isolation, frame	XIV	X					
	Initial			10	∞	---	---	MΩ
	End-of-life			1 0	∞	---	---	MΩ
P1,P3	Primary inductance (parallel)	XV	S	1 33	6.50	1.33	6 50	H
P2,P4	Primary Q	XV	S	1 00	---	1 00	---	---
R1,R2	Linearity	XVI	S	-0 0025	+0 0025	-0 0025	+0 0025	Ratio

NOTES:
1 (X) parameters shall be 100 percent tested during module acceptance tests
2 (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

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3 5.7 Holddown holes The module shall have no additional requirements for location or quantity of holddown holes.

3 5 8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/128-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3 6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3 7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I

3 8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (6.2).

4 QUALITY ASSURANCE PROVISIONS

4 1 Quality assurance The quality assurance provisions of MIL-M-28787 shall apply

4 2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4 2 1 Test conditions A and C For 100 percent acceptance testing, the transformer ratio, test condition A, and phase shift (full primary to full secondary), test condition C, may be checked at 30 V, 9 6 kHz, only

4 3 Test equipment The following test equipment is necessary to conduct acceptance tests for this module Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 designator	No required (maximum per test figure)
Sine wave generator	Item G01	1
Power amplifier	Item A01	1
Ratio transformer	Item F05	1
Power sine wave generator	Item G07	1
Ratio transformer	Item F02	1
Phase angle voltmeter	Item Z01	1
Digital, multifunction meter	Item M02	1
Isolation transformer	Item E05	1
Inductance bridge	Item L01	1
Counter-timer	Item T01	1
Sine wave generator	Item G03	1
AC voltmeter	Item W04	2
Capacitance bridge	Item C01	1
Distortion analyzer	Item B01	1
Ohmmeter	Item R01	1
Megohm bridge	Item R06	1
Resistors		
36 kΩ ±1%, 1/8 W		2
10Ω ±1%, 1/4 W		1
10Ω ±0 1%, 1/8 W		1
100Ω ±0 1%, 1/8 W		1
1 5 kΩ ±1%, 1/8 W		1
1 kΩ ±1%, 1/8 W		1

All resistors shall be metal film unless otherwise specified

4 4 Test procedures The module shall be tested as specified in table I

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4 4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4 4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions If test conditions L or N are part of a sequence of tests, the order of testing shall be as follows L, N where applicable, followed by the remaining tests in the sequence

4.5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4 4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through K, P, and R, shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2

4.5 1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2

4.5.2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be nonoperating during the test Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4 5 3 Life test The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests A, B, C, E, and G shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical tests A, B, C, E, G, L, and M shall be performed in the order specified in 4 4 2 using the 25° ±5°C ambient end-of-life requirements of table I

5 PACKAGING

5 1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6 1 Definitions Definitions are as specified in MIL-STD-1331

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (3.8).

6 3 Cross reference of classification. Part number M28787/128-1 superseded part number M28787/128. The two parts are interchangeable.

Preparing activity
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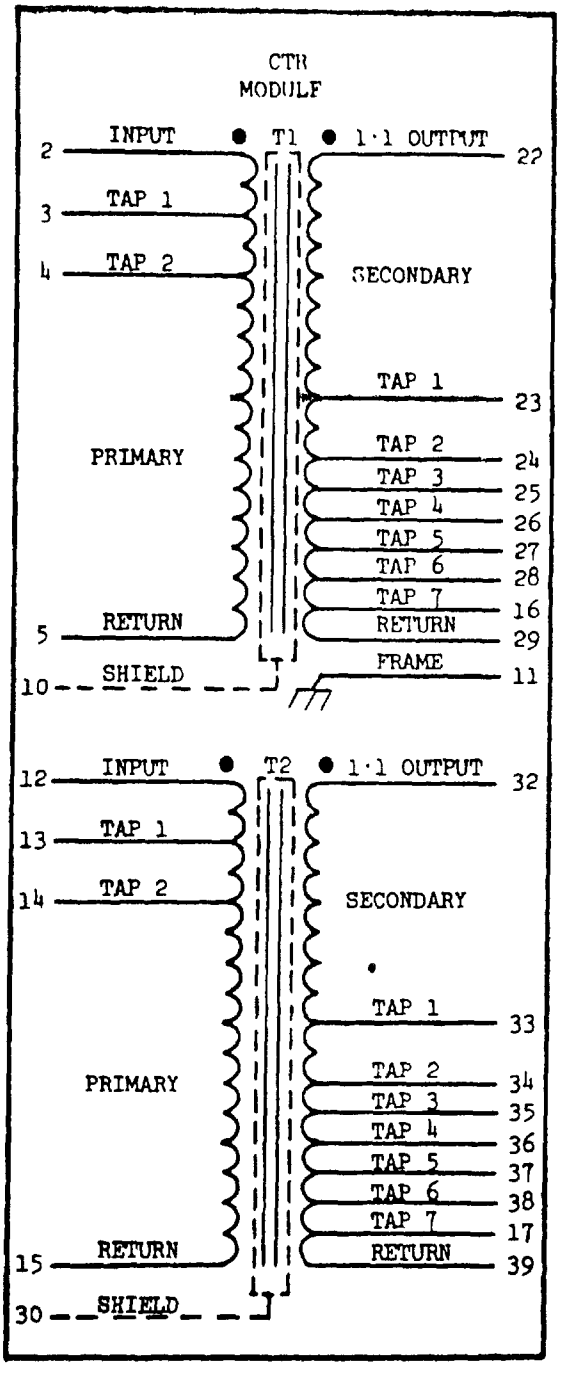
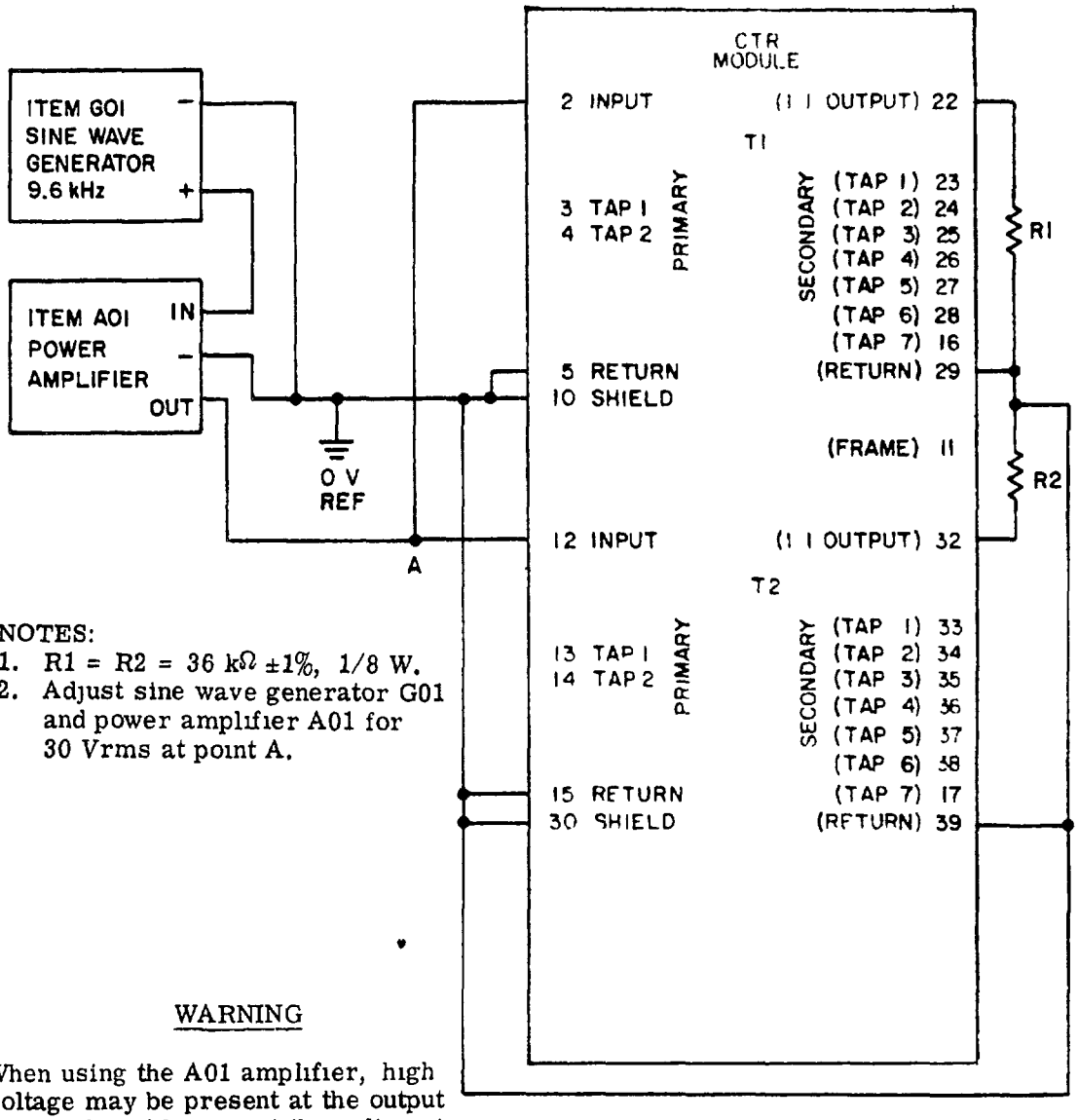


FIGURE 1. CTR function diagram, transformer.

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NOTES:
1. R1 = R2 = 36 kΩ ±1%, 1/8 W.
2. Adjust sine wave generator G01 and power amplifier A01 for 30 Vrms at point A.

WARNING

When using the A01 amplifier, high voltage may be present at the output terminals. Always set the voltage to zero before connecting or disconnecting the test leads.

FIGURE 2. Operating temperature, component temperature, and life-test circuit.

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TABLE II Transformer ratio test.

Test condition	Parameter	Connect				Voltage at A (V_A) ±1% (Vrms)	Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin		Min	Max		
A1	T1 Transformer ratio	2	5,29	22	10	1 0	1:0 9975	1:1.0025	Ratio	4
A2	T1	2	5,29	22	10	30.0				
A3	T2	12	15,39	32	30	1 0				
A4	T2 Transformer ratio	12	15,39	32	30	30.0	1:0.9975	1:1.0025	Ratio	4

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TABLE III Tap accuracy test.

Test condition	Parameter	Connect				Voltage at A (V_A) ±1% (Vrms)	Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin		Min	Max		
B1	T1 Tap 1 accuracy (primary)	2	5,29	3	10	1.0	1:0.974025	1:0.975975	Ratio	3
B2	T1 Tap 1 accuracy (primary)	↓	↓	3	↓	30.0	1:0.974025	1:0.975975	↓	↓
B3	T1 Tap 2 accuracy (primary)	↓	↓	4	↓	1.0	1:0.94905	1:0.95095	↓	↓
B4	T1 Tap 2 accuracy (primary)	2	5,29	4	10	30.0	1:0.94905	1:0.95095	↓	↓
B5	T2 Tap 1 accuracy (primary)	12	15,39	13	30	1.0	1:0.974025	1:0.975975	↓	↓
B6	T2 Tap 1 accuracy (primary)	↓	↓	13	↓	30 0	1·0.974025	1:0.975975	↓	↓
B7	T2 Tap 2 accuracy (primary)	↓	↓	14	↓	1.0	1:0.94905	1:0.95095	↓	↓
B8	T2 Tap 2 accuracy (primary)	12	15,39	14	30	30 0	1 0.94905	1:0.95095	↓	↓
B9	T1 Tap 1 accuracy (secondary)	22	5,29	23	10	1.0	1:0.4995	1:0.5005	↓	↓
B10	T1 ↓	22	5,29	23	10	30.0	↓	↓	↓	↓
B11	T2 ↓	32	15,39	33	30	1.0	↓	↓	↓	↓
B12	T2 Tap 1 accuracy (secondary)	32	15,39	33	30	30.0	1·0 4995	1·0.5005	↓	↓
B13	T1 Tap 2 accuracy (secondary)	22	5,29	24	10	1.0	1:0.283716	1:0.284284	↓	↓
B14	T1 ↓	22	5,29	24	10	30.0	↓	↓	↓	↓
B15	T2 ↓	32	15,39	34	30	1.0	↓	↓	↓	↓
B16	T2 Tap 2 accuracy (secondary)	32	15,39	34	30	30.0	1:0.283716	1:0.284284	↓	↓
B17	T1 Tap 3 accuracy (secondary)	22	5,29	25	10	1 0	1:0.228771	1:0.229229	↓	↓
B18	T1 ↓	22	5,29	25	10	30.0	↓	↓	↓	↓
B19	T2 ↓	32	15,39	35	30	1.0	↓	↓	↓	↓
B20	T2 Tap 3 accuracy (secondary)	32	15,39	35	30	30.0	1·0 228771	1:0.229229	↓	↓
B21	T1 Tap 4 accuracy (secondary)	22	5,29	26	10	1.0	1:0.178821	1:0.179179	↓	↓
B22	T1 ↓	22	5,29	26	10	30.0	↓	↓	↓	↓
B23	T2 ↓	32	15,39	36	30	1.0	↓	↓	↓	↓
B24	T2 Tap 4 accuracy (secondary)	32	15,39	36	30	30.0	1:0.178821	1:0.179179	Ratio	3

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TABLE III Tap accuracy test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Voltage at A (V_A) ±1% (Vrms)	Initial limits 25° ± 5°C Ambient Min	Initial limits 25° ± 5°C Ambient Max	Unit	Test figure
B25	T1 Tap 5 accuracy (secondary)	22	5,29	27	10	1 0	1:0.0999	1 0 1001	Ratio	3
B26	T1 ↓	22	5,29	27	10	30 0	↓	↓	↓	↓
B27	T2 ↓	32	15,39	37	30	1.0	↓	↓	↓	↓
B28	T2 Tap 5 accuracy (secondary)	32	15,39	37	30	30 0	1.0 0999	1·0.1001	↓	↓
B29	T1 Tap 6 accuracy (secondary)	22	5,29	28	10	1.0	1:0.017855	1·0.017945	↓	↓
B30	T1 ↓	22	5,29	28	10	30 0	↓	↓	↓	↓
B31	T2 ↓	32	15,39	38	30	1 0	↓	↓	↓	↓
B32	T2 Tap 6 accuracy (secondary)	32	15,39	38	30	30.0	1:0 017855	1·0 017945	↓	↓
B33	T1 Tap 7 accuracy (secondary)	22	5,29	16	10	1 0	1:0.015561	1:0.015639	↓	↓
B34	T1 ↓	22	5,29	16	10	30.0	↓	↓	↓	↓
B35	T2 ↓	32	15,39	17	30	1.0	↓	↓	↓	↓
B36	T2 Tap 7 accuracy (secondary)	32	15,39	17	30	30 0	1 0 015561	1·0 015639	Ratio	3

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TABLE IV Phase shift (full primary to full secondary) check.

Test condition	Parameter	Connect				Voltage at A (V_A) ±1% (Vrms)	Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin		Min	Max		
C1	T1 Phase shift	2	5,29	22	10	1.0	-0.25	+0.25	Degree	4
C2	T1	2	5,29	22	10	30.0				
C3	T2	12	15,39	32	30	1.0				
C4	T2 Phase shift	12	15,39	32	30	30.0	-0.25	+0.25	Degree	4

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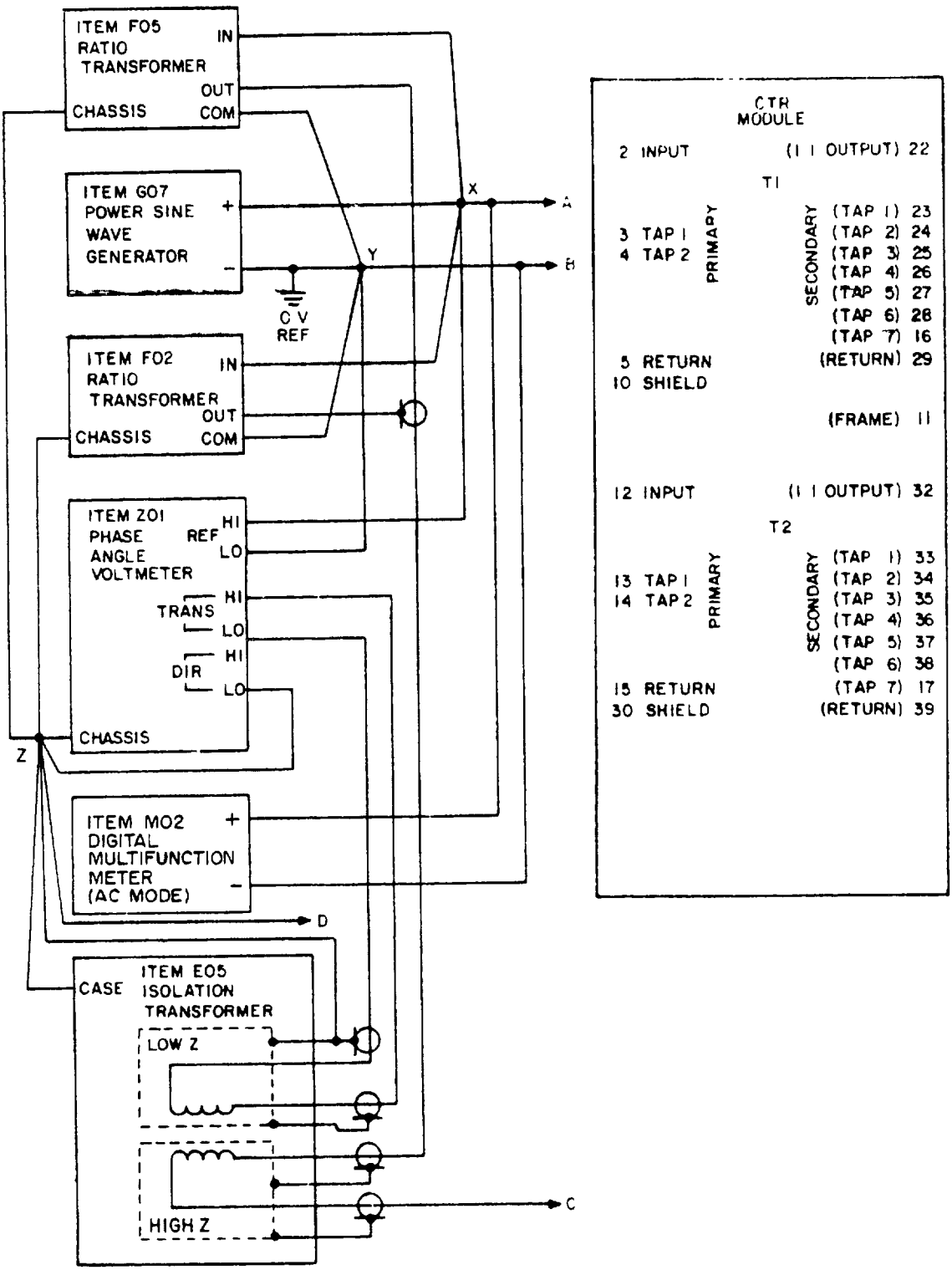


FIGURE 3. Tap accuracy test circuit for table III.

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NOTES:
1. Frequency = 9.6 kHz.
2. Test procedure:
   (a) After calibrating the phase angle voltmeter (item Z01) and making test circuit connections, set the voltage (V_A) and frequency at point A as specified. Set the function selector switch of item Z01 to the zero degree position and null the signal by adjusting the ratio transformer (item F05).
   (b) The shield shall be driven by ratio transformer (item F02) which shall be set at the same ratio as the nulling ratio transformer (item F05). A null is obtained by simultaneously adjusting both ratio transformers
3. All connections made at points X, Y, and Z shall terminate at one point as shown.
4. Disconnect item M02 before performing the tests.
5. With item G07 set to 1.0 Vrms, it may not be possible to set the reference adjust on item Z01. If this condition occurs, set the reference adjust as close as possible (maximum adjust) and continue with the test.

FIGURE 3. Tap accuracy test circuit for table III - Continued.

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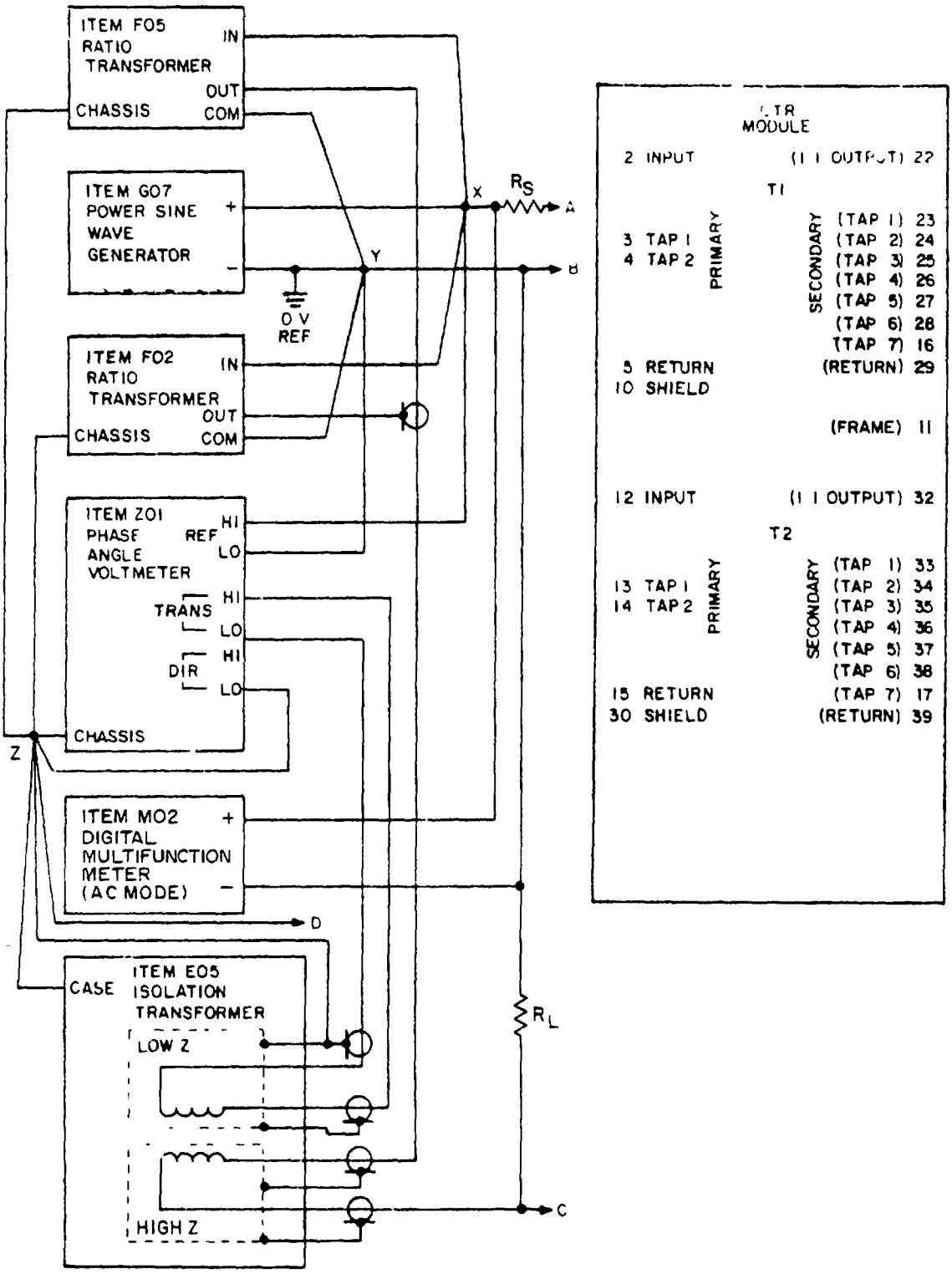


FIGURE 4. Transformer ratio and phase shift (full primary to full secondary) check test circuit for tables II and IV.

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NOTES:

1. Frequency = 9.6 kHz.
2. R_L = 36 kΩ ±1%, 1/8 W.
3. R_S = 10Ω ±1%, 1/4 W for test conditions A1-A4. R_S = 0Ω for test conditions C1-C4.
4. All connections made at points X, Y, and Z shall terminate at one point as shown.
5. Disconnect item M02 before performing the tests.
6. Transformer ratio test procedure:

   (a) After calibrating the phase angle voltmeter (item Z01) and making test circuit connections, set the voltage (V_A) and frequency at point A as specified. Set the function selector switch of item Z01 to the zero degree position and null the signal by adjusting the ratio transformer (F05).
   (b) The shield shall be driven by ratio transformer (item F02) which shall be set at the same ratio as the nulling ratio transformer (item F05). A null is obtained by simultaneously adjusting both ratio transformers.
   (c) The value of the ratio shall be within the limits specified.

7. Phase shift test procedure:

   (a) Perform test procedures 6(a) and 6(b) above.
   (b) With the in-phase signal nulled, set the function selector switch of item Z01 to the 90 degree position and read the quadrature voltage (V_Q') on item Z01.
   (c) Calculate the true value of quadrature voltage (V_Q) as follows.

   $V_Q = K (V_Q')$ where K = 4 when using isolation transformer item E05.

   (d) The values calculated in (c) above shall be as follows

   $V_Q \leq 4.37$ mV for test conditions C1 and C3
   $V_Q \leq 131$ mV for test conditions C2 and C4.

   This verifies a phase angle of ±0.25 degree.
   (e) The phase angle (Ø) may be calculated as follows:

   $$\emptyset = \tan^{-1} \frac{V_Q}{V_A}$$

   where V_A is the voltage across the ratio transformer times the ratio at null.

8. With item G07 set to 1.0 Vrms, it may not be possible to set the reference adjust on item Z01. If this occurs, set the reference adjust as close as possible (maximum adjust) and continue with test.

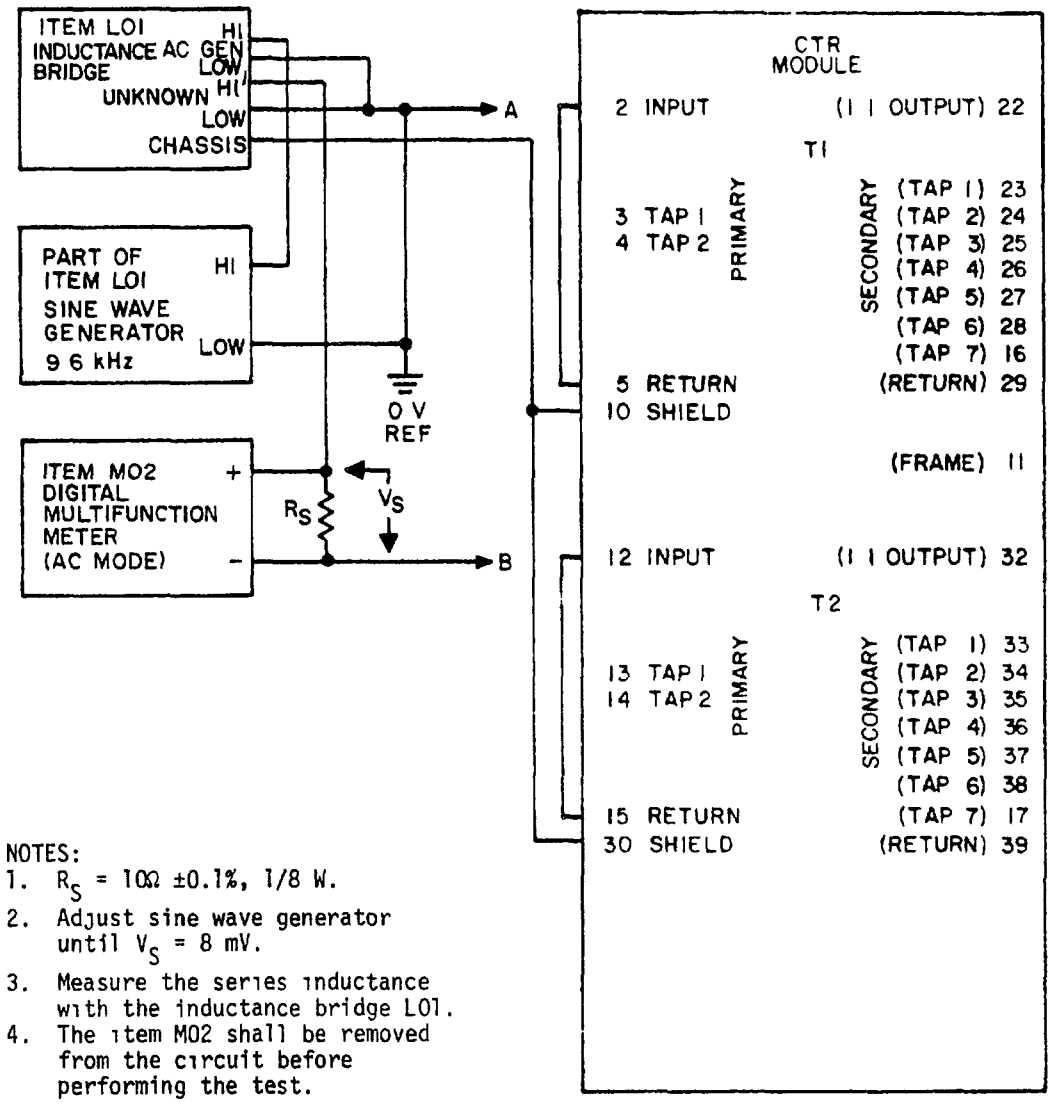
FIGURE 4. Transformer ratio and phase shift (full primary to full secondary) check test circuit for tables II and IV - Continued.

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TABLE V. Leakage inductance (reflected to secondary) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Leakage inductance (reflected to secondary)	29	22	0	0.003	H	5
D2	Leakage inductance (reflected to secondary)	39	32	0	0.003	H	5

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NOTES:
1. R_S = 10Ω ±0.1%, 1/8 W.
2. Adjust sine wave generator until V_S = 8 mV.
3. Measure the series inductance with the inductance bridge LOI.
4. The item MO2 shall be removed from the circuit before performing the test.

FIGURE 5. Leakage inductance (reflected to secondary) test circuit for table V.

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TABLE VI Primary open circuit resistance and reactance check.

Test condition	Parameter	Connect		MO2 Voltage ±1% (E_{in}) (Vrms)	Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin		Min	Max		
E1	Primary open circuit resistance and reactance	2	5,29	1.0	80,000 + j80,000	∞	Ω	6
E2	↓	2	5,29	30	↓	↓	↓	↓
E3	↓	12	15,39	1.0	↓	↓	↓	↓
E4	Primary open circuit resistance and reactance	12	15,39	30	80,000 + j80,000	∞	Ω	6

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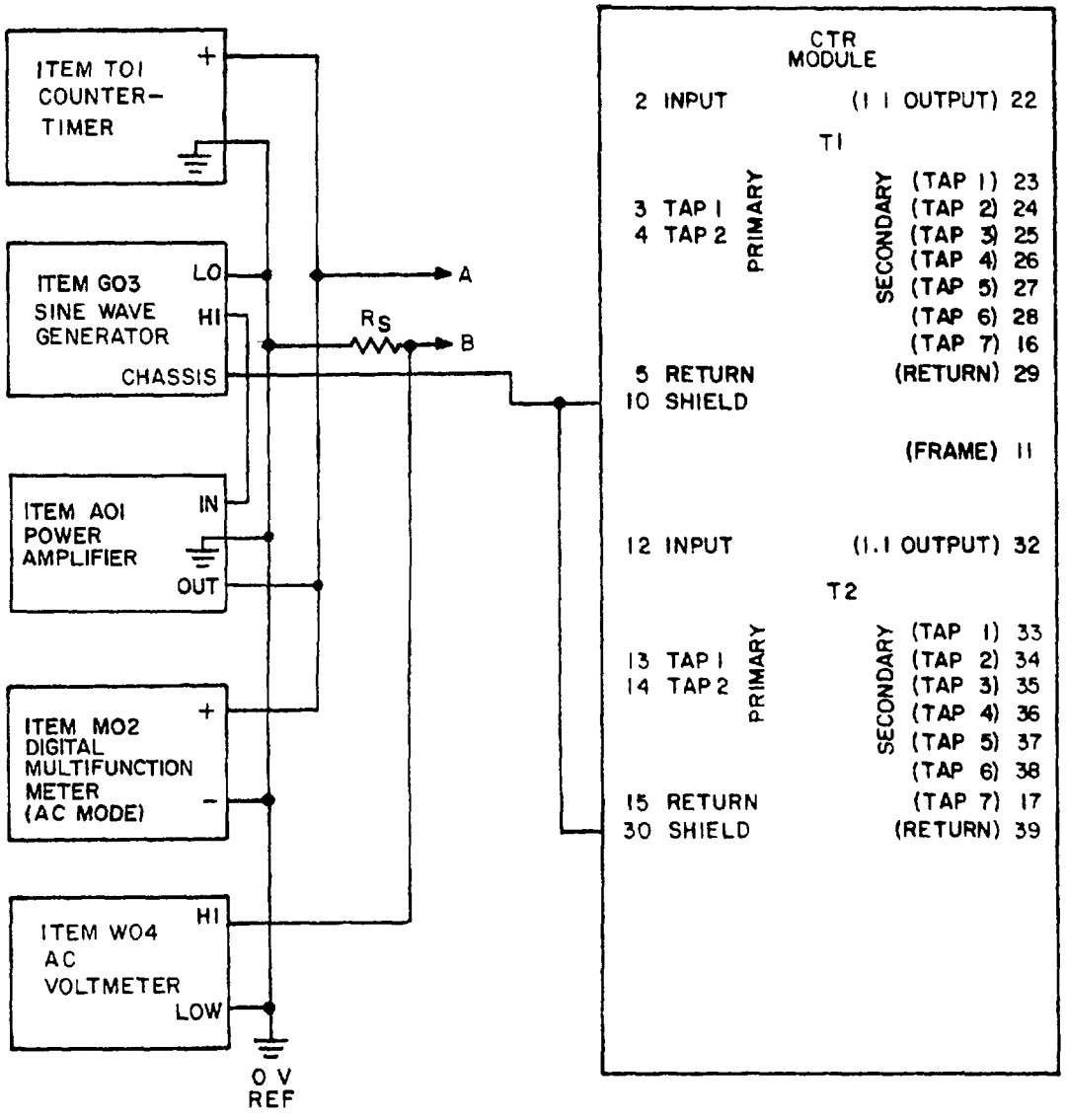


FIGURE 6. Primary open circuit resistance and reactance check test circuit for table VI.

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NOTES:

1. R_S = 100Ω ±0.1%, 1/8 W, nonınductive. (This resıstor must be physıcally located wıthın 2 ınches (50.8 mm) of the module under test.)
2. The following procedure shall be followed for each test condıtıon to verıfy that the prımary open cırcuıt resistance and reactance ıs a mınımum of 80,000 + j 80,000 ohms:

   (a) Connect the test cırcuit and adjust ıtems G03/A01 for the applıcable ıtem M02 voltage ($E_{ın}$) specifıed ın table VI.
   (b) Adjust the frequency of ıtem G03 from 9.6 kHz until ıtem W04 ındıcates a mınımum Vrms (null). Frequency at null shall be greater than 9.6 kHz (it wıll probably be less than 22 kHz, however, this ıs not a requırement). After the null has been established readjust ıtems G03/A01 for the correct voltage level.
   (c) Record the null voltage as ındıcated on ıtem W04 as V_S. The value of V_S shall be as follows:

   $V_S \leqq 1.25$ mVrms for E_{in} = 1.0 Vrms
   $V_S \leqq 37.5$ mVrms for E_{in} = 30 Vrms

   Thıs verıfıes the real part of the ımpedance to be greater than 80,000 ın accordance wıth the followıng relatıonshıp

   $$\text{Resıstance} = \left(\frac{R_S}{V_S}\right) E_{in}$$

   where R_S = 100 ohms and V_S and E_{in} are ın volts

   (d) Adjust the frequency of ıtem G03 to 9.6 kHz and readjust ıtems G03/A01 for the correct voltage ıf necessary.
   (e) Record the voltage ındıcated on ıtem W04 as V_S'.
   (f) Calculate X_L from the followıng equatıon:

   $$X_L = \frac{E_{in}}{\sqrt{\left(\frac{V_S'}{100}\right)^2 - \left(\frac{V_S}{100}\right)^2}}$$

   where E_{in}, V_S, and V_S' are ın Vrms.
   The calculated value of X_L shall be greater than 80,000 ohms.

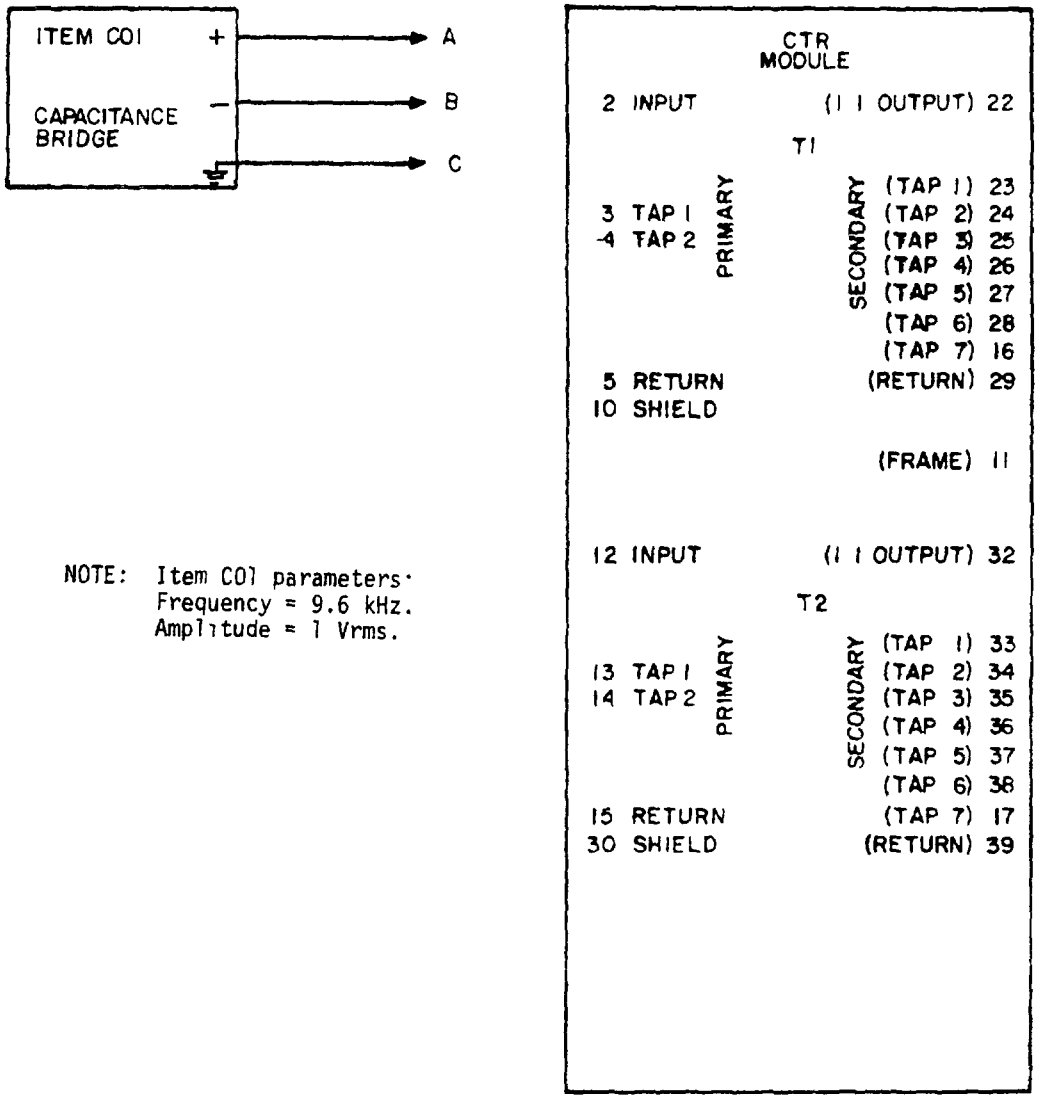
FIGURE 6. Primary open circuit resistance and reactance check test circuit for table VI - Continued.

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TABLE VII. Capacitance (primary to secondary) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	Capacitance (primary to secondary)	2,5	22,29	10	0	50	pF	7
F2	Capacitance (primary to secondary)	12,15	32,39	30	0	50	pF	7

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NOTE: Item C01 parameters:
Frequency = 9.6 kHz.
Amplitude = 1 Vrms.

FIGURE 7. Capacitance (primary to secondary) test circuit for table VII.

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TABLE VIII. DC resistance (primary and secondary).

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	DC resistance (primary)	2	5	18	33	Ω	8
G2	DC resistance (secondary)	22	29	↓	↓	↓	↓
G3	DC resistance (primary)	12	15	↓	↓	↓	↓
G4	DC resistance (secondary)	32	39	18	33	Ω	8

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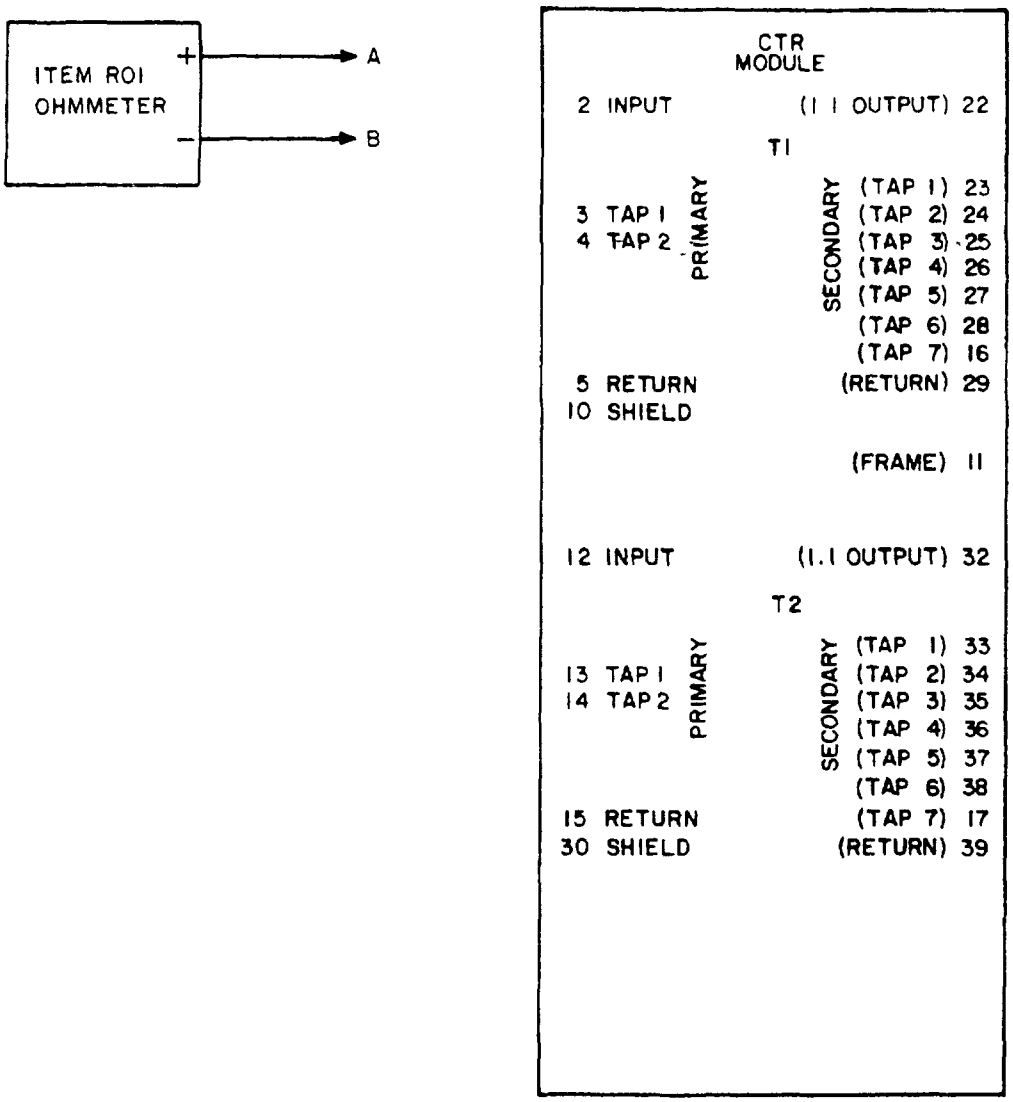


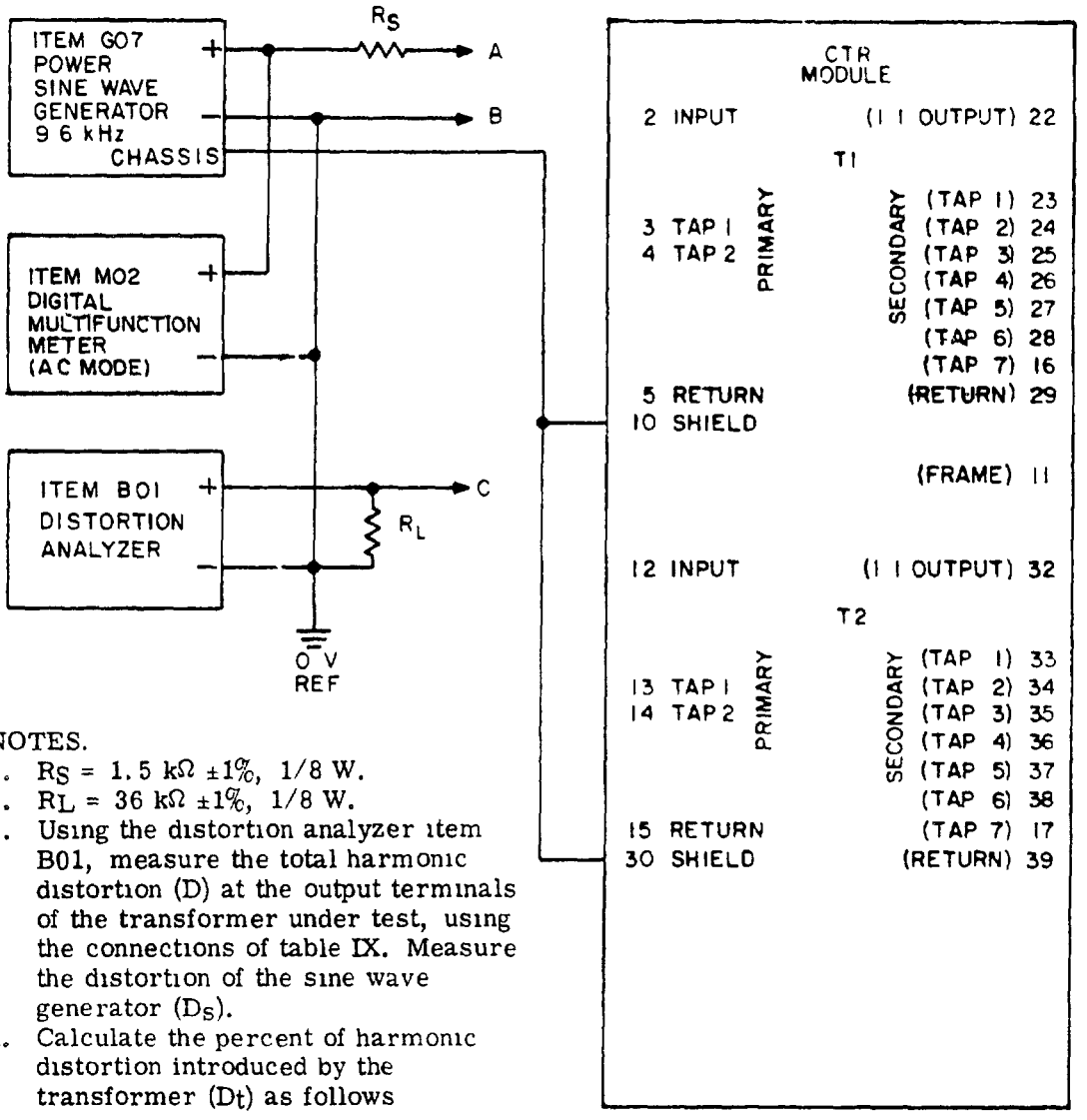
FIGURE 8. DC resistance (primary and secondary) test circuit for table VIII.

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TABLE IX. Total harmonic distortion check.

Test condition	Parameter	Connect			Voltage at $D_s \pm 1\%$ (Vrms)	Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin		Min	Max		
H1	Total harmonic distortion	2	5,29	22	1.0	0	1.0	Percent	9
H2	↓	2	5,29	22	30.0	↓	↓	↓	↓
H3	↓	12	15,39	32	1.0	↓	↓	↓	↓
H4	Total harmonic distortion	12	15,39	32	30.0	0	1.0	Percent	9

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NOTES.
1. R_S = 1.5 kΩ ±1%, 1/8 W.
2. R_L = 36 kΩ ±1%, 1/8 W.
3. Using the distortion analyzer item B01, measure the total harmonic distortion (D) at the output terminals of the transformer under test, using the connections of table IX. Measure the distortion of the sine wave generator (D_S).
4. Calculate the percent of harmonic distortion introduced by the transformer (D_t) as follows

$$D_t\ (\%) = \left(\sqrt{D^2 - D_S^2}\right)100$$

For this formula to be applicable, both D_S and D must be less than 1%.

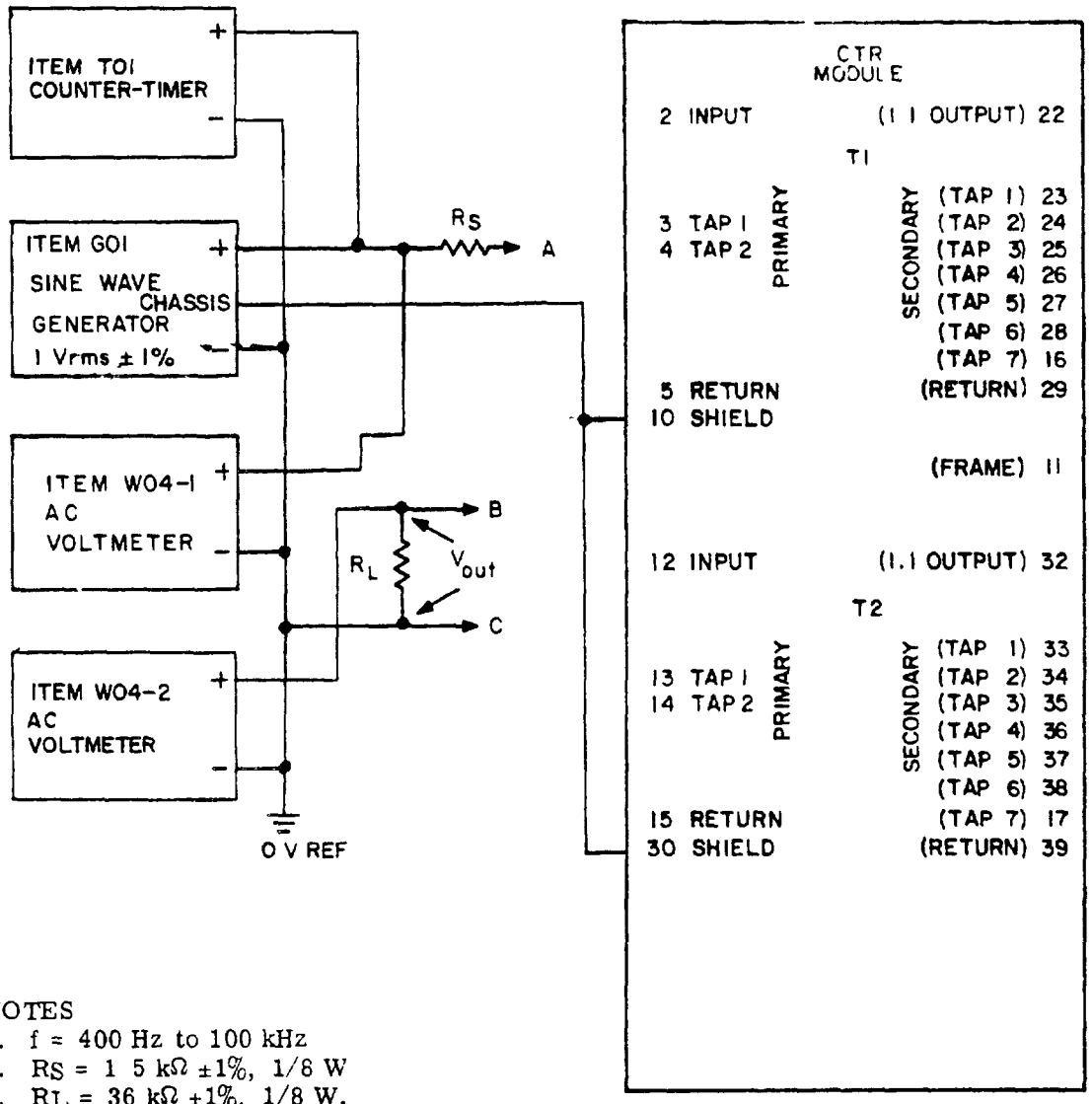
FIGURE 9. Total harmonic distortion check test circuit for table IX.

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TABLE X. Frequency response check.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Frequency response	2	22	5,29	-3	+3	dB	10
J2	Frequency response	12	32	15,39	-3	+3	dB	10

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NOTES
1. f = 400 Hz to 100 kHz
2. R_S = 1 5 kΩ ±1%, 1/8 W
3. R_L = 36 kΩ ±1%, 1/8 W.
4. With the input at 1 Vrms, 9 6 kHz, measure the output voltage (V_r). Vary the frequency over the range of 400 Hz to 100 kHz while maintaining the input voltage at 1 Vrms. Measure the output voltage (V_{out}) at frequency intervals sufficiently small to detect any output perturbation. Over the specified frequency range the voltage ratio $\left(\frac{V_{out}}{V_r}\right)$ shall be 0.707 minimum and 1.414 maximum. This verifies the frequency response of 0 ±3 dB.
5. Frequency response in dB may be calculated as follows.

$$dB = 20 \log \left(\frac{V_{out}}{V_r}\right)$$

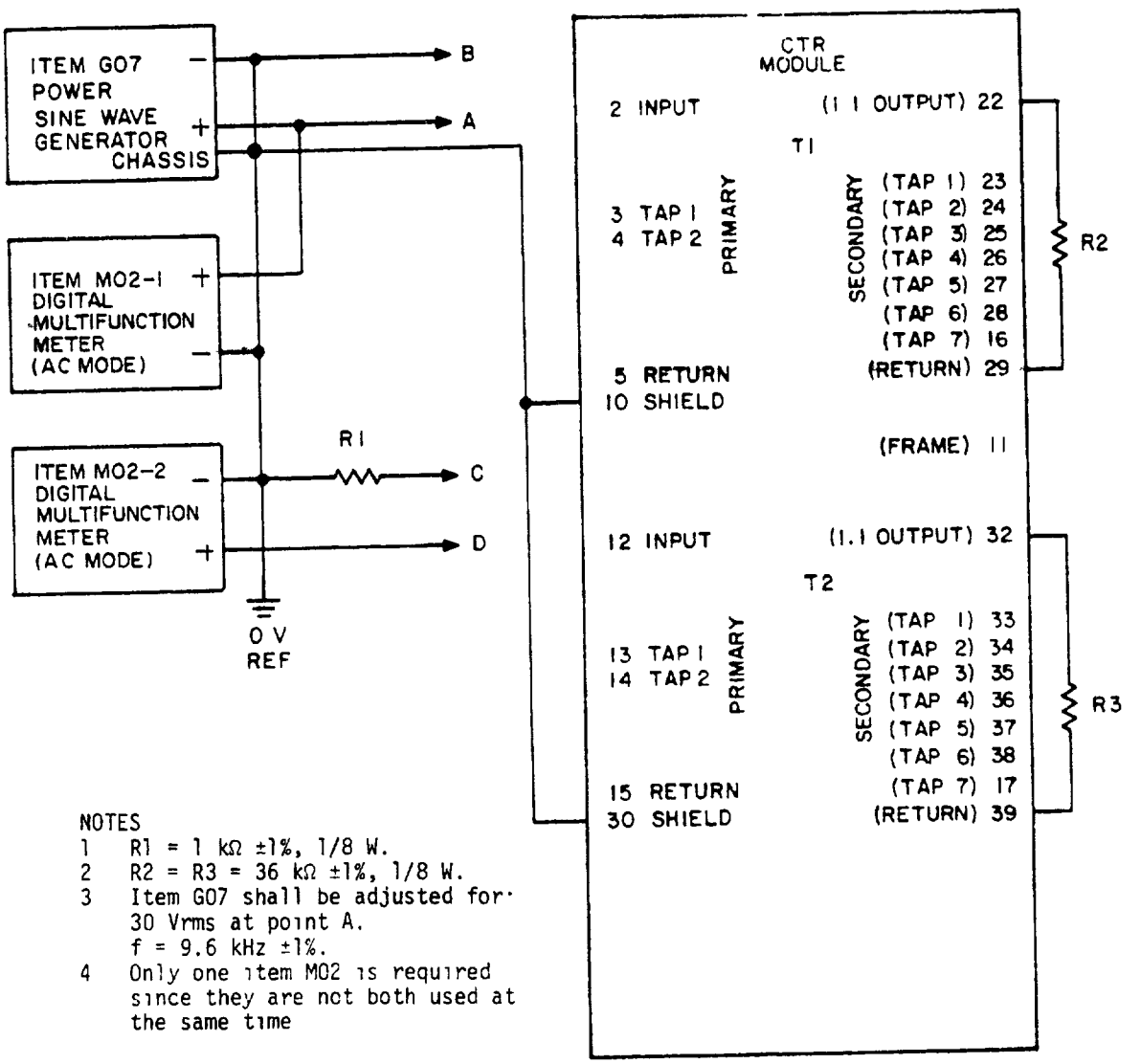
FIGURE 10. Frequency response check test circuit for table X.

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TABLE XI. Electromagnetic coupling.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	T1 Electromagnetic coupling	2	5,15, 29,39	12	32	0	0.1	Vrms	11
K2	T2 Electromagnetic coupling	12	5,15, 29,39	2	22	0	0.1	Vrms	11

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NOTES
1 R1 = 1 kΩ ±1%, 1/8 W.
2 R2 = R3 = 36 kΩ ±1%, 1/8 W.
3 Item G07 shall be adjusted for 30 Vrms at point A.
f = 9.6 kHz ±1%.
4 Only one item M02 is required since they are not both used at the same time

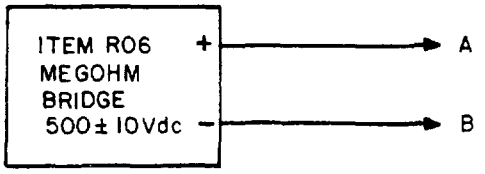
FIGURE 11. Electromagnetic coupling test circuit for table XI.

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TABLE XII. Integrity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Integrity	2-5,10-17,22-30,32-39	1,6-9,18-21,31,40	10	∞	MΩ	12
L2		1	6-9,18-21,31,40				
L3		6	7-9,18-21,31,40				
L4		7	8,9,18-21,31,40				
L5		8	9,18-21,31,40				
L6		9	18-21,31,40				
L7		18	19-21,31,40				
L8		19	20,21,31,40				
L9		20	21,31,40				
L10		21	31,40				
L11	Integrity	31	40	10	∞	MΩ	12

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WARNING.

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

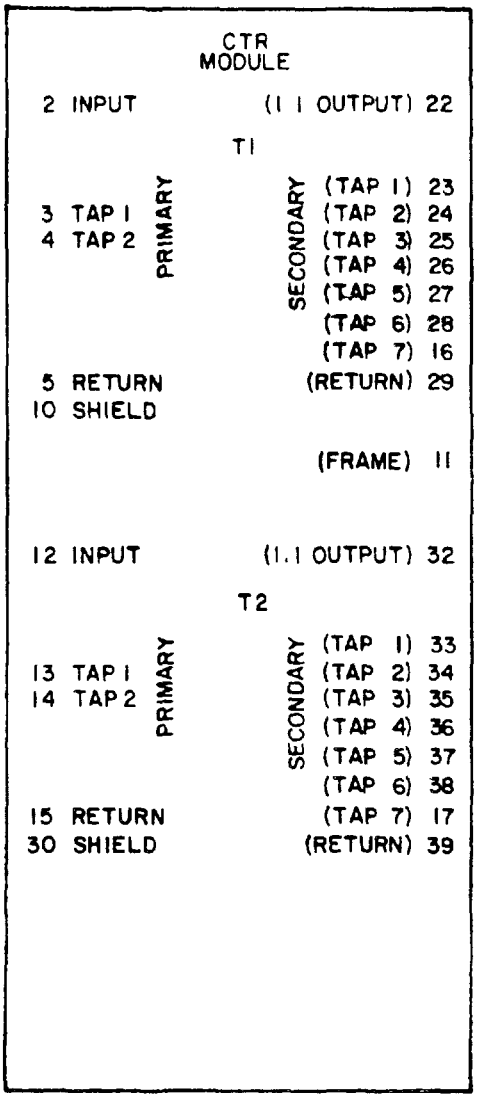


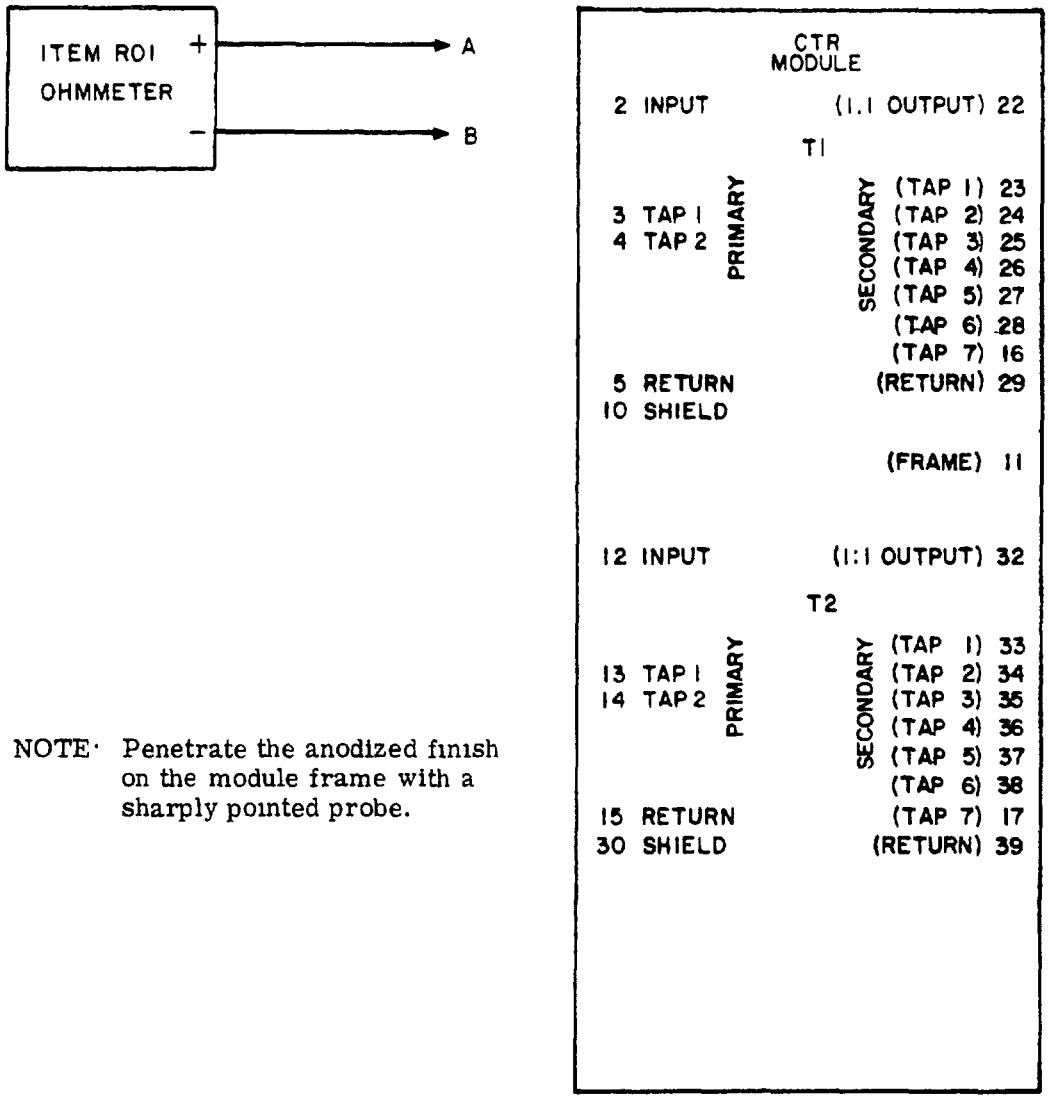
FIGURE 12. Integrity test circuit for table XII.

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TABLE XIII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M	Continuity	11	Frame	0	0.1	Ω	13

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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe.

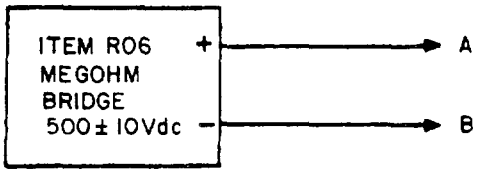
FIGURE 13. Continuity test circuit for table XIII.

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TABLE XIV. Isolation (frame) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N	Isolation, frame	11	All other pins	10	∞	MΩ	14

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

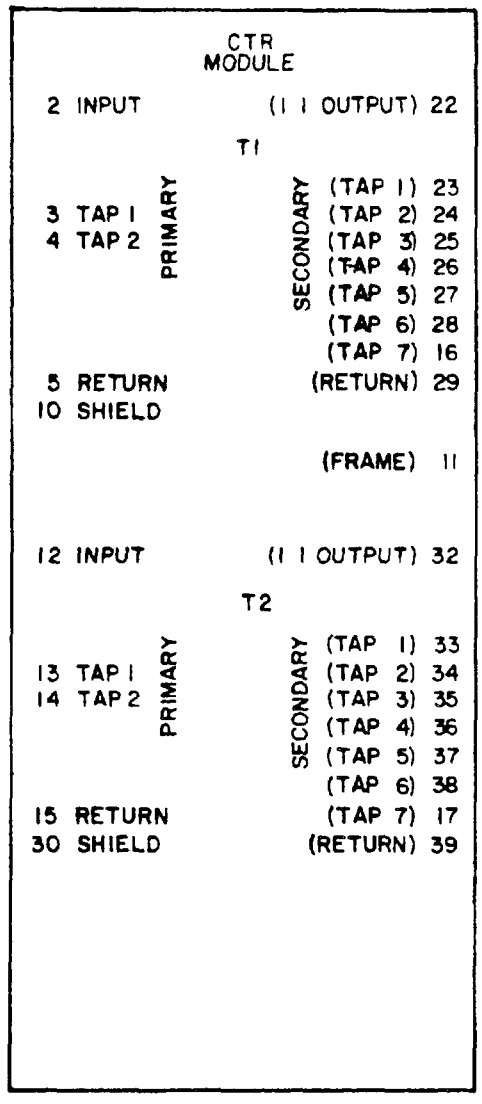


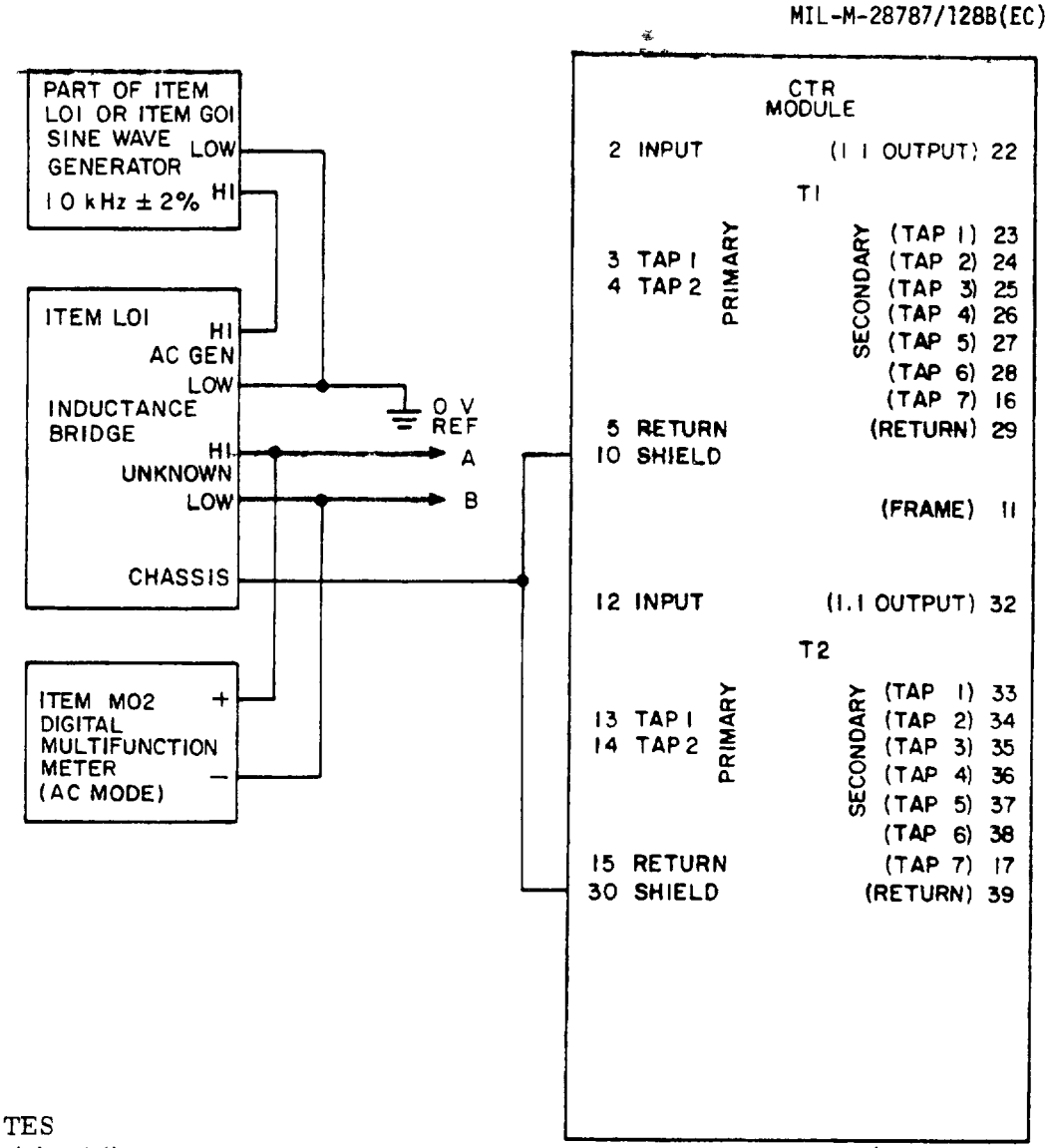
FIGURE 14 Isolation (frame) test circuit for table XIV.

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TABLE XV. Primary inductance (parallel) and primary Q test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	T1 Primary inductance (parallel)	2	5	1.33	6.50	H	15
P2	T1 Primary Q	2	5	1.00	---	---	↓
P3	T2 Primary inductance (parallel)	12	15	1.33	6.50	H	↓
P4	T2 Primary Q	12	15	1.00	---	---	15

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NOTES

1. Adjust the sine wave generator until item M02 reads 1.0 Vrms ±1% with A and B connected and the bridge approximately nulled.
2. Disconnect item M02 before making final readings.
3. If the correct voltage and/or frequency cannot be obtained using item L01, sine wave generator, use item G01.
4. Due to the low Q of some transformers being measured, the inductance bridge will have to be set to measure series inductance (L_S) and the results converted to an equivalent parallel inductance. The relationship is as follows:

$$L_P = \left(\frac{1 + Q^2}{Q^2}\right) L_S$$

Where Q is the Q obtained when measuring series inductance. The calculated L_P value and the series inductance Q shall meet the table XV limits.

FIGURE 15. Primary inductance (parallel) and primary Q test circuit for table XV.

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TABLE XVI. Linearity test

Test condition	Parameter	Calculation	Initial limits 25° ±5°C ambient		Unit	Test figure
			Min	Max		
R1	T1 Linearity	Test condition A1 - Test condition A2	-0 0025	+0 0025	Ratio	---
R2	T2 Linearity	Test condition A3 - Test condition A4	-0 0025	+0.0025	Ratio	---

NOTE: Test condition R is a calculation from the measurement data obtained in test condition A. The difference obtained from the indicated calculation shall meet the limits of test conditions R1 and R2 This test condition verifies a transformation ratio linearity (when the transformer primary goes from 1.0 to 30 0 Vac) of 0 25 percent.

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